

In the examples of cell complexes given so far, the closure of each cell is a subcomplex, and more generally the closure of any collection of cells is a subcomplex. Most naturally arising cell structures have this property, but it need not hold in general. For example, if we start with S^1 with its minimal cell structure and attach to this a 2-cell by a map $S^1 \rightarrow S^1$ whose image is a nontrivial subarc of S^1 , then the closure of the 2-cell is not a subcomplex since it contains only a part of the 1-cell.

Operations on Spaces

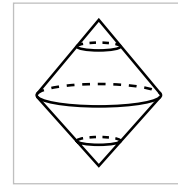
Cell complexes have a very nice mixture of rigidity and flexibility, with enough rigidity to allow many arguments to proceed in a combinatorial cell-by-cell fashion and enough flexibility to allow many natural constructions to be performed on them. Here are some of those constructions.

Products. If X and Y are cell complexes, then $X \times Y$ has the structure of a cell complex with cells the products $e_\alpha^m \times e_\beta^n$ where e_α^m ranges over the cells of X and e_β^n ranges over the cells of Y . For example, the cell structure on the torus $S^1 \times S^1$ described at the beginning of this section is obtained in this way from the standard cell structure on S^1 . For completely general CW complexes X and Y there is one small complication: The topology on $X \times Y$ as a cell complex is sometimes finer than the product topology, with more open sets than the product topology has, though the two topologies coincide if either X or Y has only finitely many cells, or if both X and Y have countably many cells. This is explained in the Appendix. In practice this subtle issue of point-set topology rarely causes problems, however.

Quotients. If (X, A) is a CW pair consisting of a cell complex X and a subcomplex A , then the quotient space X/A inherits a natural cell complex structure from X . The cells of X/A are the cells of $X - A$ plus one new 0-cell, the image of A in X/A . For a cell e_α^n of $X - A$ attached by $\varphi_\alpha: S^{n-1} \rightarrow X^{n-1}$, the attaching map for the corresponding cell in X/A is the composition $S^{n-1} \rightarrow X^{n-1} \rightarrow X^{n-1}/A^{n-1}$.

For example, if we give S^{n-1} any cell structure and build D^n from S^{n-1} by attaching an n -cell, then the quotient D^n/S^{n-1} is S^n with its usual cell structure. As another example, take X to be a closed orientable surface with the cell structure described at the beginning of this section, with a single 2-cell, and let A be the complement of this 2-cell, the 1-skeleton of X . Then X/A has a cell structure consisting of a 0-cell with a 2-cell attached, and there is only one way to attach a cell to a 0-cell, by the constant map, so X/A is S^2 .

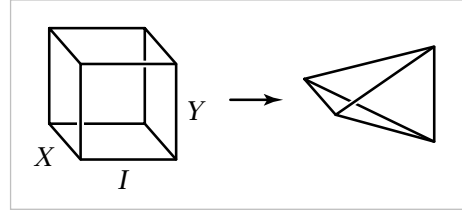
Suspension. For a space X , the **suspension** SX is the quotient of $X \times I$ obtained by collapsing $X \times \{0\}$ to one point and $X \times \{1\}$ to another point. The motivating example is $X = S^n$, when $SX = S^{n+1}$ with the two ‘suspension points’ at the north and south poles of S^{n+1} , the points $(0, \dots, 0, \pm 1)$. One can regard SX as a double cone



on X , the union of two copies of the **cone** $CX = (X \times I)/(X \times \{0\})$. If X is a CW complex, so are SX and CX as quotients of $X \times I$ with its product cell structure, I being given the standard cell structure of two 0-cells joined by a 1-cell.

Suspension becomes increasingly important the farther one goes into algebraic topology, though why this should be so is certainly not evident in advance. One especially useful property of suspension is that not only spaces but also maps can be suspended. Namely, a map $f: X \rightarrow Y$ suspends to $Sf: SX \rightarrow SY$, the quotient map of $f \times \mathbb{1}: X \times I \rightarrow Y \times I$.

Join. The cone CX is the union of all line segments joining points of X to an external vertex, and similarly the suspension SX is the union of all line segments joining points of X to two external vertices. More generally, given X and a second space Y , one can define the space of all line segments joining points in X to points in Y . This is the **join** $X * Y$, the quotient space of $X \times Y \times I$ under the identifications $(x, y_1, 0) \sim (x, y_2, 0)$ and $(x_1, y, 1) \sim (x_2, y, 1)$. Thus we are collapsing the subspace $X \times Y \times \{0\}$ to X and $X \times Y \times \{1\}$ to Y . For example, if X and Y are both closed intervals, then we are collapsing two opposite faces of a cube onto line segments so that the cube becomes a tetrahedron. In the general case, $X * Y$ contains copies of X and Y at its two ‘ends,’ and every other point (x, y, t) in $X * Y$ is on a unique line segment joining the point $x \in X \subset X * Y$ to the point $y \in Y \subset X * Y$, the segment obtained by fixing x and y and letting the coordinate t in (x, y, t) vary.



A nice way to write points of $X * Y$ is as formal linear combinations $t_1x + t_2y$ with $0 \leq t_i \leq 1$ and $t_1 + t_2 = 1$, subject to the rules $0x + 1y = y$ and $1x + 0y = x$ that correspond exactly to the identifications defining $X * Y$. In much the same way, an iterated join $X_1 * \cdots * X_n$ can be regarded as the space of formal linear combinations $t_1x_1 + \cdots + t_nx_n$ with $0 \leq t_i \leq 1$ and $t_1 + \cdots + t_n = 1$, with the convention that terms $0t_i$ can be omitted. This viewpoint makes it easy to see that the join operation is associative. A very special case that plays a central role in algebraic topology is when each X_i is just a point. For example, the join of two points is a line segment, the join of three points is a triangle, and the join of four points is a tetrahedron. The join of n points is a convex polyhedron of dimension $n - 1$ called a **simplex**. Concretely, if the n points are the n standard basis vectors for $\mathbb{R}^n$, then their join is the space $\Delta^{n-1} = \{(t_1, \dots, t_n) \in \mathbb{R}^n \mid t_1 + \cdots + t_n = 1 \text{ and } t_i \geq 0\}$.

Another interesting example is when each X_i is S^0 , two points. If we take the two points of X_i to be the two unit vectors along the i^{th} coordinate axis in $\mathbb{R}^n$, then the join $X_1 * \cdots * X_n$ is the union of 2^n copies of the simplex Δ^{n-1} , and radial projection from the origin gives a homeomorphism between $X_1 * \cdots * X_n$ and S^{n-1} .

If X and Y are CW complexes, then there is a natural CW structure on $X * Y$ having the subspaces X and Y as subcomplexes, with the remaining cells being the product cells of $X \times Y \times (0, 1)$. As usual with products, the CW topology on $X * Y$ may be weaker than the quotient of the product topology on $X \times Y \times I$.

Wedge Sum. This is a rather trivial but still quite useful operation. Given spaces X and Y with chosen points $x_0 \in X$ and $y_0 \in Y$, then the **wedge sum** $X \vee Y$ is the quotient of the disjoint union $X \amalg Y$ obtained by identifying x_0 and y_0 to a single point. For example, $S^1 \vee S^1$ is homeomorphic to the figure ‘8,’ two circles touching at a point. More generally one could form the wedge sum $\bigvee_{\alpha} X_{\alpha}$ of an arbitrary collection of spaces X_{α} by starting with the disjoint union $\bigsqcup_{\alpha} X_{\alpha}$ and identifying points $x_{\alpha} \in X_{\alpha}$ to a single point. In case the spaces X_{α} are cell complexes and the points x_{α} are 0-cells, then $\bigvee_{\alpha} X_{\alpha}$ is a cell complex since it is obtained from the cell complex $\bigsqcup_{\alpha} X_{\alpha}$ by collapsing a subcomplex to a point.

For any cell complex X , the quotient X^n/X^{n-1} is a wedge sum of n -spheres $\bigvee_{\alpha} S_{\alpha}^n$, with one sphere for each n -cell of X .

Smash Product. Like suspension, this is another construction whose importance becomes evident only later. Inside a product space $X \times Y$ there are copies of X and Y , namely $X \times \{y_0\}$ and $\{x_0\} \times Y$ for points $x_0 \in X$ and $y_0 \in Y$. These two copies of X and Y in $X \times Y$ intersect only at the point (x_0, y_0) , so their union can be identified with the wedge sum $X \vee Y$. The **smash product** $X \wedge Y$ is then defined to be the quotient $X \times Y / X \vee Y$. One can think of $X \wedge Y$ as a reduced version of $X \times Y$ obtained by collapsing away the parts that are not genuinely a product, the separate factors X and Y .

The smash product $X \wedge Y$ is a cell complex if X and Y are cell complexes with x_0 and y_0 0-cells, assuming that we give $X \times Y$ the cell-complex topology rather than the product topology in cases when these two topologies differ. For example, $S^m \wedge S^n$ has a cell structure with just two cells, of dimensions 0 and $m+n$, hence $S^m \wedge S^n = S^{m+n}$. In particular, when $m = n = 1$ we see that collapsing longitude and meridian circles of a torus to a point produces a 2-sphere.

Two Criteria for Homotopy Equivalence

Earlier in this chapter the main tool we used for constructing homotopy equivalences was the fact that a mapping cylinder deformation retracts onto its ‘target’ end. By repeated application of this fact one can often produce homotopy equivalences between rather different-looking spaces. However, this process can be a bit cumbersome in practice, so it is useful to have other techniques available as well. We will describe two commonly used methods here. The first involves collapsing certain subspaces to points, and the second involves varying the way in which the parts of a space are put together.